

Remote Sensing Image Change Caption Generation — Consolidated Report

Project overview, experiment timeline and findings, July – October 2026

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Abstract

This report consolidates the full project on remote sensing image change captioning (RSICC) for Indian imagery. It covers three parts:

- **Part A (July–August):** we built a modular captioning pipeline on the LEVIR-CC benchmark and developed it through four architectures, improving BLEU-4 from 30.3 to 33.7. Attempts to extend it beyond LEVIR-CC failed. Sequential fine-tuning on SECOND-CC caused catastrophic forgetting, and designs with large language-model decoders could not be trained to completion.
- **Part B (September):** a pilot Indian Sentinel-2 collection was captioned with off-the-shelf vision-language models, which produced unusable output.
- **Part C (September–October):** we built an India-wide dataset of 15,610 quality-screened Sentinel-2 patch pairs and evaluated whether open-weight vision-language models could label land-use change automatically. Across prompting strategies, visual assistance, model sizes and a 1,000-pair validation against human labels, the best rule was correct on at most about half of its change calls.

We conclude that neither transfer from LEVIR-CC nor automatic labelling with current vision-language models yields reliable change captions for Indian 10 m imagery, and that scaling either approach would produce a dataset of unacceptably poor quality.

1. Introduction

Remote sensing image change captioning describes, in natural language, what changed between two satellite images of the same place. Public RSICC datasets such as LEVIR-CC use sub-metre imagery of a few foreign regions. India's freely available imagery is medium-resolution (Sentinel-2, 10 m), and its landscapes are dominated by smallholder farmland, monsoon-driven seasonal cycles, haze and dense organic settlements.

The project aimed at a change captioning system for Indian imagery. It moved through three parts, each motivated by the failure of the previous one:

Part	Question	Outcome
A	Can a model trained on LEVIR-CC be improved and adapted to new data?	Improves on LEVIR-CC, but does not transfer
B	Can existing vision-language captioners caption Indian imagery directly?	No: empty, garbled or "nothing changed" output
C	Can we build an Indian dataset and label it automatically with vision-language models?	Dataset built; automatic labels too inaccurate to use

2. Project timeline

Date (2026)	Part	Work	Result
2 Jul	A	Phase 1: modular pipeline and CNN baseline on LEVIR-CC	BLEU-4 30.3, ROUGE-L 62.1
7–9 Jul	A	Phase 2: frozen RemoteCLIP encoder with difference fusion	BLEU-4 32.8; largest single gain
9–10 Jul	A	Final model: RemoteCLIP cross-attention + contrastive alignment	BLEU-4 33.7
12–28 Jul	A	Phase 6: hierarchical tile-based model	BLEU-4 33.6
Jul	A	Sequential fine-tuning on SECOND-CC (Phase 6 and Final models)	Catastrophic forgetting: LEVIR-CC BLEU-4 fell to 12–15
30 Jul – 1 Aug	A	Phases 7–8: tile-difference variants and hyperparameter changes	Marked failed; degraded
2–20 Aug	A	CodeAug / Phase 8: RemoteCLIP + Q-Former / bridge + Qwen language-model decoder	Stage-2 loss diverged to NaN; no evaluation completed
9–10 Sep	B	Pilot Indian Sentinel-2 collection; TeoChat and Qwen2.5-VL captioning	1,072 pairs kept; captions unusable
mid-Sep	C	Prototype: change heatmaps and a two-stage VLM captioner on a few sites	Captioner dismissed a real localized change; selection problems found
27 Sep	C	All-India downloader, 1,200-site registry, pilot	Resumable downloader; quality checks added
28 Sep	C	Haze screen trial on the GPU cluster	4-bit 8B model chosen for speed
28–29 Sep	C	Full dataset: download, patching, screening	15,610 accepted patch pairs
29 Sep	C	Diagnosis of failed captions; perception test of four models	Seasonal change mistaken for land-use change
30 Sep	C	Change gate (pixel and CNN features) and two-model triage trial	Strict triage found only 7 of 15 changes
1 Oct	C	Prompting with and without assistance; teammate's change-map method	Boxes halve detection; forcing prompts and maps invent change
1–2 Oct	C	1,000-pair validation; blind human review of 535 pairs	Best rule correct on at most about half of its change calls

3. Part A — LEVIR-CC models (July–August)

3.1 Pipeline. All models shared one codebase, so their results are comparable:

- the official LEVIR-CC splits
- a single vocabulary
- caption cross-entropy loss
- Adam with learning rate 1×10^{-4} and a cosine schedule
- evaluation on all 1,929 test pairs

3.2 Architectures and results (LEVIR-CC test set):

Model	Description	BLEU-4	METEOR	ROUGE-L
Phase 1	Two-stream CNN encoder + Transformer decoder	30.3	59.7	62.1
Phase 2	Frozen RemoteCLIP ViT-B/32, difference fusion	32.8	63.0	65.5
Phase 6	Tile-based RemoteCLIP with tile fusion	33.6	63.6	66.2
Final	RemoteCLIP cross-attention + contrastive alignment	33.7	63.9	66.0

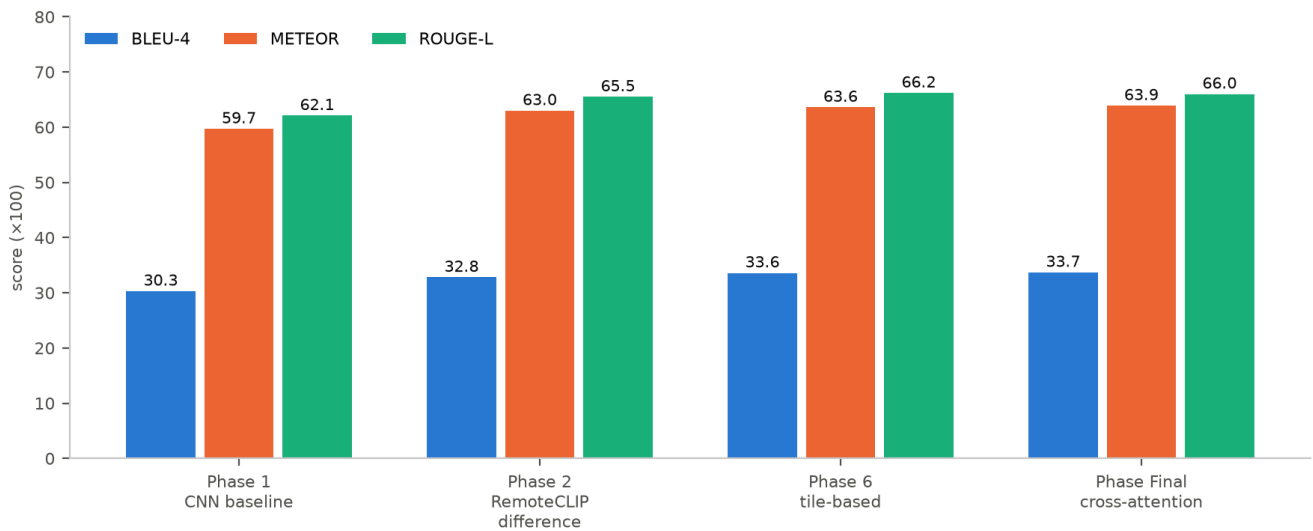


Figure 1. LEVIR-CC test scores per architecture.

A remote-sensing-pretrained encoder gave the largest single gain, and further complexity added little.

3.3 Cross-dataset extension. Sequential fine-tuning on SECOND-CC erased most of the LEVIR-CC performance (Figure 2), while SECOND-CC scores stayed low (BLEU-4 12–13). Language-model decoders (Qwen2-VL-2B, Qwen2-0.5B) were implemented, but training diverged to NaN in half precision, and no evaluation was completed.

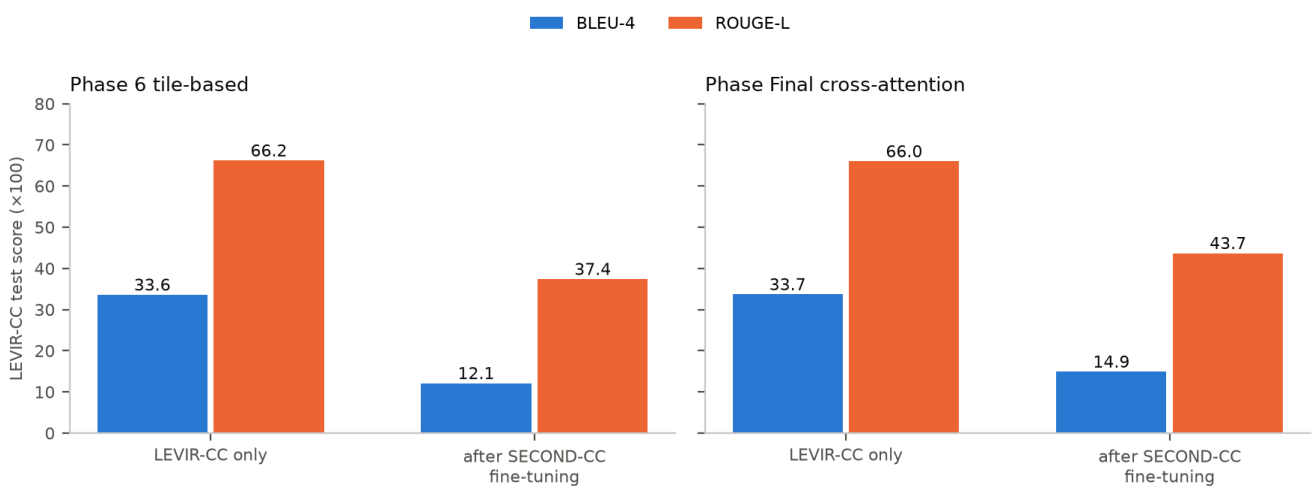


Figure 2. LEVIR-CC test scores before and after sequential fine-tuning on SECOND-CC.

4. Part B — Off-the-shelf captioners on Indian imagery (September)

A pilot Sentinel-2 collection over hand-picked Indian areas kept 1,072 of 2,126 tiles. Two pretrained captioners were run on a 30-pair sample:

- **TeoChat** answered "Nothing meaningful changed" for 27 of 30 pairs.
- **Qwen2.5-VL** produced empty or garbled output for many pairs. When it did answer, it described generic urban development (Figure 3).



Before

After

Qwen2.5-VL: "...increased urban density and expansion of buildings, particularly along the central axis, indicating rapid development. There's also visible growth in road infrastructure..."

TeoChat: "Nothing meaningful changed."

Figure 3. One Indian pair with both captioners' answers. Neither describes the specific change visible on the right.

5. Part C — India-wide dataset and automatic labelling (September–October)

5.1 Dataset. - **Sites:** 1,200 sites spread evenly over all 36 states and union territories (Figure 4). - **Imagery:** Sentinel-2 Collection-1 true colour at 10 m, with before (2019–20) and after (2025–26) images taken in matching dry-season windows. - **Quality control:** block-level cloud, snow and water limits, haze and whiteout checks, alignment checks, patch-level screening, and a haze screen run on the GPU cluster. - **Result:** 27,888 patch pairs, of which **15,610 were accepted.**

Download sites: 1200 evenly spread over 36 states/UTs

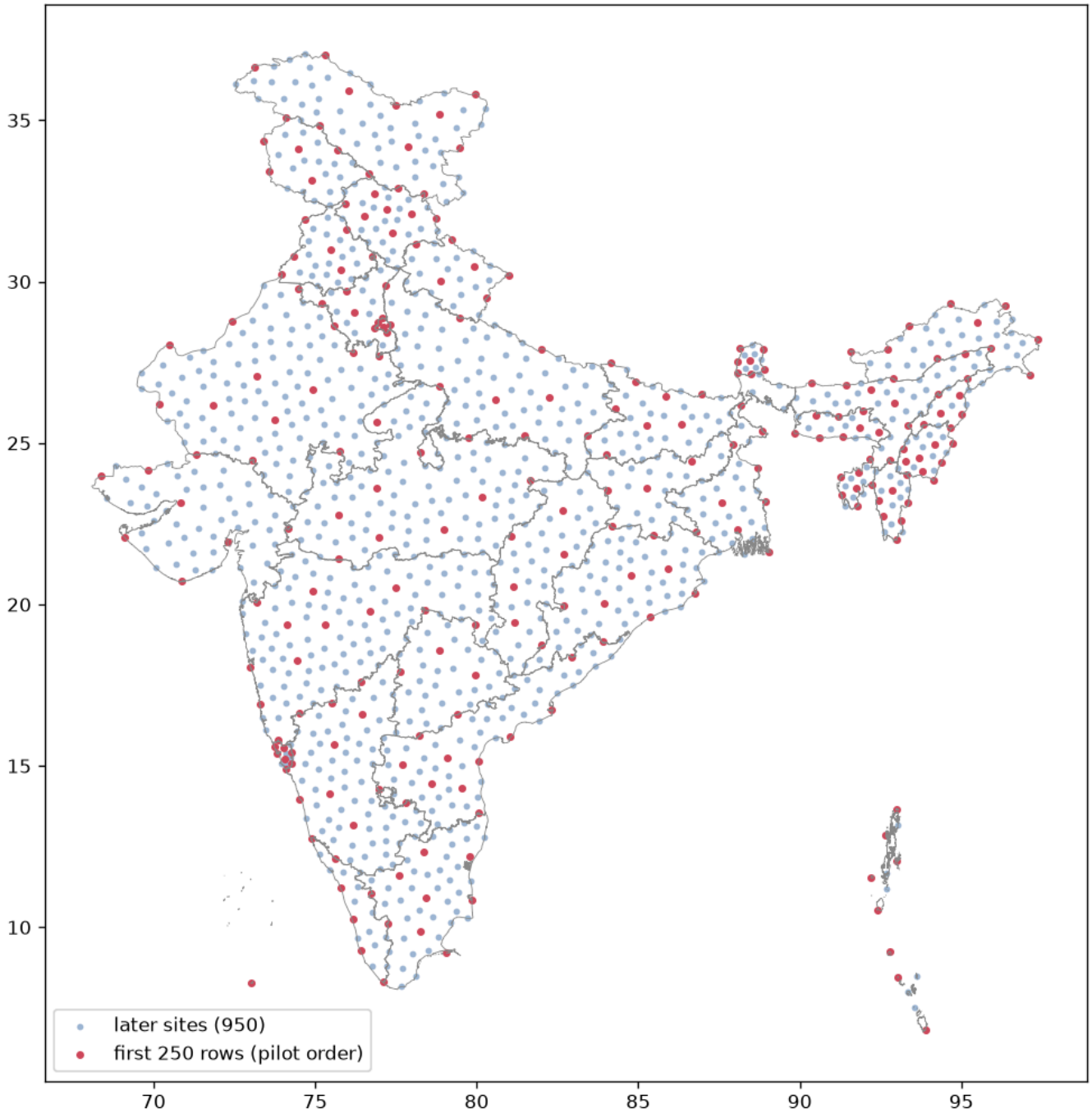


Figure 4. Site coverage across India.

5.2 Reference labels. A web-based review tool shows the before and after images with a blink view. A human annotator labelled 535 pairs blind, choosing change, seasonal, no change or can't tell. The pairs were those where the automatic labels disagreed, plus a random sample where they agreed.

5.3 Prompting experiments. - **Without assistance:** constrained, neutral and free-form prompts all shifted the balance between missed and false changes, but none removed the confusion between seasonal and lasting change. - **With assistance:** neutral region boxes, whether from pixel differences or CNN features, roughly halved detection (Figure 5). A forcing prompt invented change on 12 of 15 identical-image pairs. A rule-based change map marked a median 82% of each image as changed and led the model to report change on every pair (Figure 6).

Plain prompt: change found (correct). Human label: Change (bihar_004_r2c1)



Before



After

Pixel boxes: answer 'none' (missed)

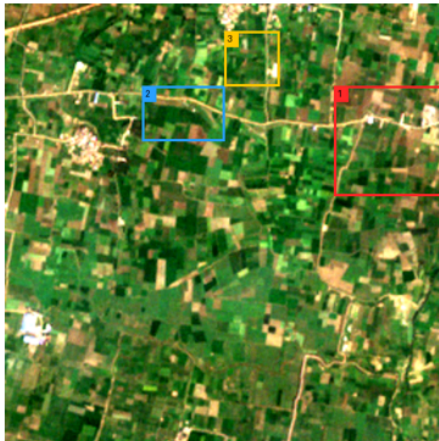


Before, pixel-difference boxes



After, pixel-difference boxes

CNN boxes: answer 'seasonal' (missed)



Before, CNN boxes



After, CNN boxes

Figure 5. A human-confirmed change, found with the plain prompt but missed once region boxes are added.

Human label: Seasonal (madhya_pradesh_071_r3c3). Before | After | change map

32B caption with map: "Large areas of bare soil became vegetated, vegetation changed across the scene, and buildings were removed in the left and bottom-left, leaving vegetation."



- bare soil -> vegetation (37.2%)
- vegetation changed (32.4%)
- vegetation -> shadow (9.6%)
- building -> vegetation (8.1%)
- building changed (2.8%)
- bare soil -> building (0.7%)
- building -> bare soil (0.6%)
- road -> vegetation (0.5%)
- vegetation -> building (0.4%)
- bare soil -> shadow (0.2%)
- road -> bare soil (0.0%)
- road -> building (0.0%)
- building -> road (0.0%)
- building -> shadow (0.0%)

Figure 6. Rule-based change map on a pair the human labelled seasonal.

5.4 1,000-pair validation. The free-form question was put to Qwen3-VL-32B and Qwen3-VL-235B on 1,002 pairs (API cost USD 1.00).

Decision rule	Human-labelled pairs (n = 495): found / precision	All pairs (n = 962): found / precision
32B	89% / 11%	96% / 26%
235B	45% / 17%	79% / 48%
32B and 235B agree	42% / 18%	78% / 51%

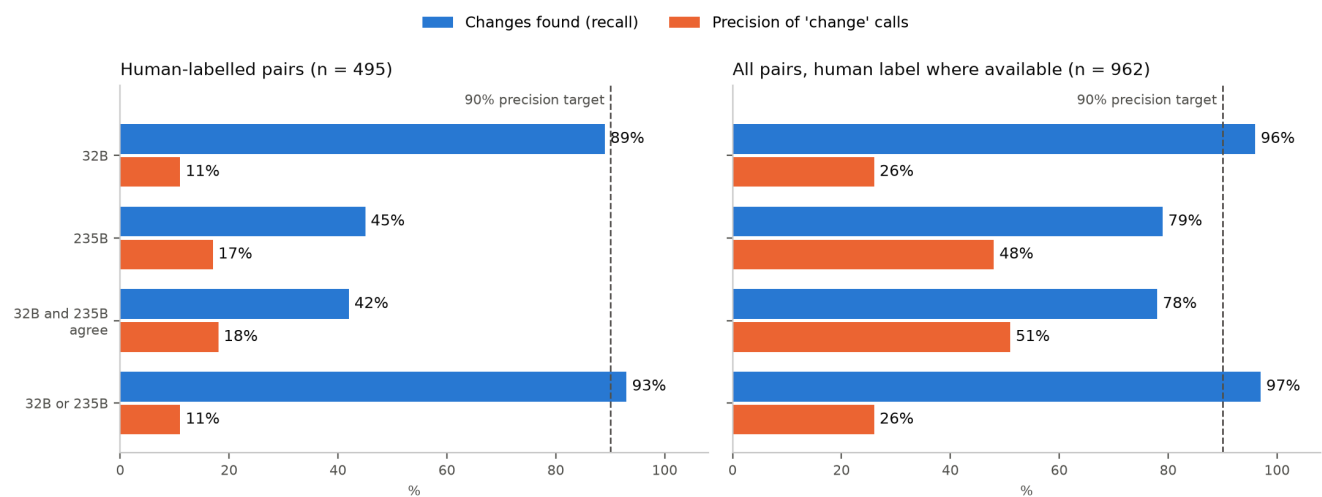


Figure 7. Changes found and precision of "change" calls for each rule, against the 90% precision needed for automatic labelling.

(a) Correct: human Change; 32B and 235B say change



(b) False alarm: human Seasonal; 32B and 235B say change



(c) Missed: human Change; 32B and 235B say seasonal



Figure 8. Validation examples with the human label as ground truth: (a) correct, (b) false alarm, (c) missed change.

Real change appears in only about one pair in seven, so even a low false-change rate produces as many false "change" labels as true ones. Automatic labelling was therefore rejected as a source for the dataset.

6. Cross-cutting findings

1. **Data is the bottleneck, not architecture.** On LEVIR-CC, architectural changes beyond a pretrained encoder brought diminishing returns. For India, no suitable labelled data existed, and none of the attempts to create it automatically produced reliable labels.

2. **Seasonal change is the central failure.** Every model, from LEVIR-CC-trained captioners to Qwen3-VL-235B, confused crop cycles, greenness, water level and haze with lasting land-use change.
3. **Hints and forced prompts are harmful.** Telling a model where or that something changed either narrows its attention or makes it invent change. This held in the early two-stage captioner, in the region-box tests and in the change-map method.
4. **Adaptation by fine-tuning is fragile.** Sequential fine-tuning destroyed source-domain performance without delivering target-domain quality.
5. **Compute shaped the project.** The institute cluster provided one older GPU at a time without bfloat16 support. It limited model size, forced aggressive quantisation, caused numerical failures and was unreachable during the final experiments.

7. Limitations

- LEVIR-CC scores use one reference caption per image, so they are comparable across our models but not with published results.
- The language-model decoder designs never completed an evaluation run.
- Off-the-shelf captioners were checked on only 30 Indian pairs, without reference captions.
- At 10 m resolution, much land-use change is at or below the limit of what can be seen.
- The human-labelled evaluation set comes from a single annotator and is weighted toward difficult pairs. The remaining reference labels carry an estimated error of about 20%.
- The prompting experiments used 40–135 pairs each, so their numbers carry wider uncertainty than the 1,000-pair validation.

8. Conclusion

Over four months we built a reproducible RSICC pipeline on LEVIR-CC and an India-wide, quality-screened Sentinel-2 dataset of 15,610 patch pairs. We tested three routes to Indian change captioning:

- transferring LEVIR-CC models
- applying off-the-shelf vision-language captioners
- labelling the new dataset automatically with open-weight vision-language models

All three failed for the same underlying reason: no available model reliably tells lasting land-use change from seasonal variation in medium-resolution Indian imagery. Scaling any of these pipelines would produce a large dataset of poor quality. Reliable change labels for this setting still require human annotation, and the dataset, tooling and evaluation protocol built here provide the base for that work.

Detailed reports: Phase 1 (LEVIR-CC and adaptation) and the Indian dataset report, provided separately.